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EAGLE

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1. Open-Source Softwares

1.1 Software Overview

Table 1.1: Comparison of open-source software for EAGLE requirements

EAGLE Requirement	PyPSA	pandapower	OpenDSS	MATPOWER	VeraGrid
Open-source	Yes	Yes	Yes	Yes	Yes
Python-native workflow	Native	Native	Via interfaces	MATLAB / Octave	Native
Programmatic automation	Yes	Yes	Yes	Yes	Yes
HV transmission modelling	Strong	Strong	Secondary focus	Strong	Strong
MV / distribution modelling	Possible	Strong	Strong	Possible	Strong
HVDC modelling	Yes	Yes	Limited / indirect	Limited	Yes
Non-linear AC power flow	Yes	Yes	Yes	Yes	Yes
DC / linear power flow	Yes	Yes	Limited	Yes	Yes
Line loading / thermal limits	Yes	Yes	Yes	Yes	Yes
Voltage constraints	Yes	Yes	Yes	Yes	Yes
Time-series analysis	Strong	Strong	Strong	MOST / additional workflow	Yes
Contingency / N-1 analysis	Yes	Yes	Possible	Additional workflow	Yes
PTDF / LODF analysis	Possible	Possible	Not core	Possible	Native
Optimal Power Flow (OPF)	Strong	Yes	Limited	Strong	Yes
Long-term planning / expansion	Strong	Limited	Limited	Additional tools	Possible
Hosting-capacity studies	Custom methodology	Custom methodology	Strong for distribution	Custom methodology	Nodal functionality
Large-scale network studies	Strong	Yes	Distribution-oriented	Strong	Yes
PyPSA-Eur compatibility	Native	Conversion required	Conversion required	Conversion required	PyPSA import
CIM / CGMES interoperability	External workflow	Partial / conversion	External workflow	External workflow	Supported
Geographical network data	Coordinates	Geodata	Basic coordinates	Not GIS-oriented	Geographic data
GIS / map visualisation	Basic / external	Yes	External	No	Yes
Python GIS / SiteOptimizer integration	Direct	Direct	Via Python interface	Additional integration	Direct
Main strength for EAGLE	**Large-scale planning and optimisation**	**Detailed electrical analysis**	**Distribution and hosting capacity**	**Power flow and OPF reference**	**Transmission analysis and interoperability**

Green indicates that the capability is directly supported and well aligned with the corresponding EAGLE requirement. Orange indicates that the capability is available with limitations, through additional modules or interfaces, or is not a primary focus of the software. Red indicates that the capability is not natively supported s poorly aligned with the requirement.
sting capacity is not necessarily a single software function. In several frameworks it must be implemented as a methodology combining power-flow or optimal-power-flow calculations with voltage, thermal and operational constraints.

1.2 Software Description

1.2.1 PyPSA

Developer / organisation: PyPSA community. Development and maintenance are currently led by the Department of Digital Transformation in Energy Systems at the Technical University of Berlin.

Access:

- Official website: https://pypsa.org/
- Documentation: https://docs.pypsa.org/
- Source repository: https://github.com/PyPSA/PyPSA

Description: PyPSA (Python for Power System Analysis) is an open-source Python framework for modelling, simulating and optimising electrical and multi-energy systems. It is particularly oriented towards large-scale transmission systems, renewable integration, storage, sector coupling and operational and investment optimisation over long time series.

Main capabilities:

- Static non-linear AC power flow.
- Linearised power flow.
- Economic dispatch.
- Linear optimal power flow.
- Security-constrained optimal power flow.
- Unit commitment.
- Generation and storage capacity optimisation.
- Transmission expansion optimisation.
- Multi-period and multi-investment-period optimisation.
- Renewable generation modelling.
- Storage and hydro modelling.
- Sector coupling and multi-carrier energy systems.
- Network and result visualisation.

Network components:

- Buses.
- AC transmission lines.
- Transformers.
- Controllable links and HVDC connections.

- Generators.
- Loads.
- Storage units.
- Energy stores.
- Shunt impedances.
- Multi-carrier conversion technologies.

Power-system analysis:

- Full non-linear AC power flow using Newton–Raphson.
- Linearised AC/DC network analysis.
- Network-constrained economic dispatch.
- Linear optimal power flow.
- Security-constrained linear optimal power flow considering network contingencies.
- Operational optimisation over multiple time steps.
- Capacity-expansion and transmission-expansion optimisation.

Data handling and automation: PyPSA is implemented in Python and stores network components and time-dependent information in structures closely integrated with pandas. This facilitates automated workflows, modification of network parameters, batch simulations, processing of long time series and integration with external Python data-analysis libraries.

GIS / spatial capabilities: Network buses can contain geographical coordinates and PyPSA provides network plotting and exploration functionality. Its Python-based architecture facilitates integration with GeoPandas and other GIS libraries. The PyPSA ecosystem also includes PyPSA-Eur, which provides geographically referenced European energy-system models.

Input / output formats: PyPSA networks can be imported and exported using machine-readable formats including NetCDF, CSV and HDF5-based workflows depending on the functionality used. Network and result tables can also be directly accessed and processed through Python and pandas.

Main advantages for EAGLE:

- Native Python environment suitable for automated EAGLE workflows.
- Direct compatibility with PyPSA-Eur, currently considered as a candidate source for the initial Spanish transmission-network model.
- Designed for large transmission networks and long time series.
- Strong support for renewable generation and storage.
- Network constraints can be included directly in optimisation.
- Suitable for studying future electricity-system scenarios.

- Supports both operational and infrastructure-investment optimisation.
- Results can be processed programmatically for integration with the spatial SiteOptimizer workflow.

Main limitations for EAGLE:

- Many optimisation functions rely on linearised network formulations rather than full non-linear AC equations.
- It is primarily oriented towards system-level and planning studies rather than highly detailed equipment-level electrical analysis.
- Detailed short-circuit and protection studies are not its primary purpose.
- Additional validation may be required when using reconstructed or estimated network parameters from PyPSA-Eur.

Potential application in EAGLE: PyPSA could serve as the main simulation and optimisation framework of the EAGLE electrical-network model. A Spanish network extracted from PyPSA-Eur could be enriched with official Spanish data and used to calculate network flows, line loading, congestion and available network capacity under different demand and generation scenarios. Its optimisation capabilities are particularly relevant for evaluating future data-centre demand, renewable integration and possible network reinforcement scenarios.

1.2.2 pandapower

Developer / organisation: Joint development involving the Energy Management and Power System Operation research group at the University of Kassel and the Grid Planning and Grid Operation Division at Fraunhofer IEE.

Access:

- Official website: https://www.pandapower.org/
- Documentation: https://pandapower.readthedocs.io/
- Source repository: https://github.com/e2nIEE/pandapower

Description: pandapower is an open-source Python framework for modelling, analysis and optimisation of electrical power systems. It combines detailed element-based electrical models with a pandas-based data structure and is designed for highly automated static and quasi-static power-system studies.

Main capabilities:

- AC power flow.
- DC power flow.
- Optimal power flow.
- State estimation.
- Short-circuit calculation.

- Topological network analysis.
- Time-series simulations.
- Controller-based simulations.
- Network plotting and geographical visualisation.
- Conversion from and to several external network formats.

Network components:

- Buses.
- AC lines.
- DC lines.
- Two-winding and three-winding transformers.
- Loads.
- Generators.
- Static generators.
- External grids.
- Switches.
- Shunts.
- Storage systems.
- Impedances and other electrical elements.

Power-system analysis:

- Newton–Raphson AC power flow.
- DC power flow.
- Optimal power flow.
- State estimation and bad-data analysis.
- IEC 60909 short-circuit calculations.
- Network topology analysis using graph-based methods.
- Quasi-static time-series simulation.

Data handling and automation: Network elements and simulation results are stored in pandas DataFrames, such as buses, lines, transformers, loads and their corresponding result tables. This structure is particularly convenient for automated data processing, filtering, modification, validation and integration with other Python libraries.

GIS / spatial capabilities: pandapower supports geographical information associated with network elements and provides plotting functionality. Its Python environment also facilitates

integration with external GIS libraries for more advanced spatial analysis.

Input / output formats: Networks can be stored and exchanged through several formats and converters, including JSON, Excel and MATPOWER/PYPOWER-compatible representations. Network tables can also be directly manipulated and exported using pandas.

Main advantages for EAGLE:

- Native Python integration.
- Detailed and validated electrical component models.
- Convenient pandas-based representation of network elements and results.
- Detailed calculation of bus voltages, branch flows, loading and electrical losses.
- State-estimation and short-circuit functionality if required in later EAGLE developments.
- NetworkX-based topology analysis can be useful for validating reconstructed network connectivity.
- Standard line and transformer libraries can help complete missing electrical parameters.
- Suitable for automated batch and time-series studies.

Main limitations for EAGLE:

- Long-term generation and transmission capacity-expansion optimisation is not its main design focus.
- Energy-system planning and sector coupling are less central than in PyPSA.
- Direct integration with PyPSA-Eur would require conversion of the network representation.
- For very large multi-period optimisation problems, a framework specifically designed for energy-system optimisation may be more convenient.

Potential application in EAGLE: pandapower could be used as a detailed electrical-analysis and validation framework for the EAGLE network. It is particularly suitable for checking AC power-flow results, voltage limits, line loading, electrical losses and network connectivity. It could therefore either serve as the main electrical simulation engine or complement a PyPSA-based planning model with more detailed electrical-network analyses.

1.2.3 OpenDSS

Developer / organisation: Electric Power Research Institute (EPRI).

Access:

- Official documentation: https://opendss.epri.com/
- Official downloads: https://sourceforge.net/projects/electricdss/
- Python interface ecosystem: https://dss-extensions.org/

- OpenDSSDirect.py: https://github.com/dss-extensions/OpenDSSDirect.py

Description: OpenDSS (Open Distribution System Simulator) is an open-source electrical system simulation tool developed primarily for utility distribution systems and distributed-energy-resource integration. It supports detailed multi-phase network modelling and is particularly oriented towards distribution planning, DER studies and quasi-static time-series simulation.

Main capabilities:

- Multi-phase power-flow analysis.
- Unbalanced distribution-network modelling.
- Quasi-static time-series simulation.
- Daily and yearly load-shape simulation.
- Distributed-generation and DER analysis.
- Voltage-regulation studies.
- Harmonic analysis.
- Fault and short-circuit studies.
- Energy-metering and loss analysis.
- Distribution planning studies.

Network components:

- Buses and nodes.
- Multi-phase lines and cables.
- Transformers.
- Loads.
- Generators.
- PV systems.
- Storage systems.
- Capacitors.
- Reactors.
- Voltage regulators.
- Protection and switching devices.

Power-system analysis:

- Multi-phase AC circuit analysis.
- Unbalanced power flow.

- Long quasi-static time-series simulations.
- Voltage and overload analysis.
- Harmonic studies.
- Fault and short-circuit calculations.
- Detailed analysis of distributed generation and storage.

Data handling and automation: OpenDSS uses its own text-based scripting language and can be automated through external programming interfaces. Python integration is possible through projects such as DSS-Python and OpenDSSDirect.py. This enables automated simulations and integration with the wider Python ecosystem, although these interfaces should be distinguished from the official EPRI implementation.

GIS / spatial capabilities: Geographical coordinates can be associated with buses and network elements, but GIS-based transmission-system modelling is not the primary focus of OpenDSS. Advanced spatial processing would normally be performed using external tools.

Input / output formats: The native OpenDSS model is based primarily on text-based DSS files and commands. Simulation results can be exported for subsequent processing, and Python interfaces can be used to retrieve results directly from the simulation engine.

Main advantages for EAGLE:

- Mature framework for detailed electrical simulation.
- Strong capabilities for time-series analysis.
- Detailed modelling of distributed generation, storage and voltage-control devices.
- Particularly suitable if EAGLE is later extended towards distribution-level hosting-capacity studies.
- Python automation is possible through the DSS-Extensions ecosystem.

Main limitations for EAGLE:

- Primarily designed for distribution and subtransmission systems rather than large European transmission-system planning.
- Large-scale transmission optimisation is not its main purpose.
- Less directly compatible with the PyPSA-Eur data ecosystem.
- Integration into a fully Python-native EAGLE workflow requires an additional interface such as DSS-Python or OpenDSSDirect.py.
- Its detailed unbalanced distribution capabilities may exceed the requirements of the initial EAGLE transmission-network model.

Potential application in EAGLE: OpenDSS would be particularly valuable if EAGLE is extended from the high-voltage transmission system towards detailed distribution-network hosting-capacity analysis. For the initial national transmission-level model, its main

strengths are less closely aligned with the current EAGLE requirements than those of transmission-oriented Python frameworks.

1.2.4 MATPOWER

Developer / organisation: MATPOWER project, originally developed at Cornell University and maintained by the MATPOWER development community.

Access:

- Official website: https://matpower.org/
- Documentation: https://matpower.org/doc/
- Source repository: https://github.com/MATPOWER/matpower

Description: MATPOWER is a free and open-source package for steady-state electric power-system simulation and optimisation in MATLAB and GNU Octave. It is widely used in research and education and provides a compact bus-branch representation for power-flow and optimal-power-flow studies.

Main capabilities:

- AC power flow.
- DC power flow.
- Continuation power flow.
- AC optimal power flow.
- DC optimal power flow.
- Economic dispatch and market-related analyses.
- Extensible mathematical optimisation framework.
- Multi-period optimisation through MOST.
- Large collection of benchmark network cases.

Network components:

- Buses.
- Branches representing lines and transformers.
- Generators.
- Loads represented through bus demand.
- Generator costs.
- Additional components through extensions and the newer MATPOWER architecture.

Power-system analysis:

- Newton-based AC power flow.

- DC power flow.
- Continuation power flow.
- AC optimal power flow.
- DC optimal power flow.
- Multi-period scheduling and optimisation through MOST.

Data handling and automation: MATPOWER uses MATLAB/Octave structures and M-files to represent electrical networks. This provides a compact and highly reproducible representation and makes automated studies straightforward within MATLAB or GNU Octave. Integration with a Python-centred workflow, however, normally requires conversion or an additional interface.

GIS / spatial capabilities: MATPOWER is primarily an electrical and mathematical simulation framework and does not provide a GIS-oriented network data model as a core feature. Geographical processing would generally be performed externally.

Input / output formats: The standard MATPOWER case format stores bus, generator, branch and cost information in MATLAB structures. The format is widely supported by other open-source power-system tools, making MATPOWER cases useful for network interchange and benchmark studies.

Main advantages for EAGLE:

- Mature and extensively used power-flow and OPF framework.
- Simple and transparent mathematical network representation.
- Large collection of standard test systems.
- Useful as an independent reference for validating EAGLE power-flow or optimisation results.
- MATPOWER case format is supported by several other power-system tools.
- GNU Octave provides a fully open-source execution alternative to MATLAB.

Main limitations for EAGLE:

- Not Python-native.
- Limited native GIS and spatial-analysis capabilities.
- Less direct integration with the current PyPSA-Eur data workflow.
- The traditional bus-branch data model is less convenient than element-based models when handling detailed equipment information.
- Long-term energy-system and renewable-planning functionality requires additional tools such as MOST or custom implementations.

Potential application in EAGLE: MATPOWER could provide an independent reference implementation for validating power-flow and optimal-power-flow calculations performed by the main EAGLE framework. Its widespread case format may also be useful for ex-

changing network models with other tools. However, adopting it as the central EAGLE framework would require additional integration with the Python and geospatial workflow.

1.2.5 VeraGrid (formerly GridCal)

Developer / organisation: VeraGrid open-source project, led by Santiago Peñate Vera. VeraGrid is the current continuation of the project previously known as GridCal.

Access:

- Documentation: https://veragrid.readthedocs.io/
- Source repository: https://github.com/SanPen/VeraGrid
- Python package: https://pypi.org/project/VeraGrid/

Description: VeraGrid is an open-source, cross-platform power-system planning and simulation platform written in Python. It combines a graphical user interface with a Python calculation engine and provides a broad range of power-system analysis, optimisation and data-conversion capabilities. It is the current continuation of GridCal.

Main capabilities:

- AC/DC multi-grid power flow.
- Three-phase unbalanced power flow.
- Linear optimal power flow.
- PTDF and LODF linear analysis.
- Net transfer capacity calculations.
- Contingency analysis.
- Stochastic power flow.
- Short-circuit calculations.
- Continuation power flow.
- Time-series simulations.
- Investment analysis.
- RMS and EMT simulation capabilities.
- Graphical network editing and visualisation.
- Network model validation and repair tools.

Network components:

- Buses and substations.
- AC transmission lines.
- HVDC connections.

- Transformers.
- Loads.
- Generators.
- Batteries and storage elements.
- Shunts.
- Switches and other network devices.
- Detailed line and equipment templates.

Power-system analysis:

- Non-linear AC/DC power flow.
- Linearised network analysis.
- Linear optimal power flow.
- Contingency and N-1 analysis.
- PTDF and LODF calculation.
- Optimal net transfer capacity analysis.
- Stochastic power flow.
- Short-circuit and continuation power-flow analysis.
- Time-series simulation for multiple study types.

Data handling and automation: VeraGrid is implemented in Python and separates its calculation engine from its graphical interface. The VeraGridEngine package can therefore be used directly as a Python library for scripting, automated batch calculations and integration into external applications. A server package is also available for remote access to the calculation engine.

GIS / spatial capabilities: VeraGrid includes network schematic and substation-line map representations and can handle geographical information associated with network assets. Its Python architecture also allows integration with external geospatial-processing libraries.

Input / output formats: VeraGrid supports a broad range of power-system formats, including MATPOWER, PSS/E RAW/RAWX, CIM, CGMES, PyPSA, JSON and its native VeraGrid formats. This makes it particularly useful for exchanging and validating network models between different software ecosystems.

Main advantages for EAGLE:

- Native Python implementation.
- Graphical interface in addition to programmatic access.
- Broad range of transmission-system analysis functions.
- Contingency, PTDF, LODF and transfer-capacity analyses are directly relevant to

transmission-network constraints.

- Supports time-series simulations.
- Can import PyPSA network models.
- Strong interoperability through MATPOWER, CIM/CGMES, PSS/E and other network formats.
- Could provide an independent environment for visually inspecting and validating the EAGLE network.

Main limitations for EAGLE:

- The current project has recently transitioned from the GridCal name to VeraGrid, so older publications and documentation may use different package and class names.
- Its ecosystem and user community are smaller than those of some more established frameworks.
- Direct use of PyPSA-Eur requires an import/conversion step, even though PyPSA models are supported.
- Long-term European energy-system capacity-expansion workflows are less directly aligned with the PyPSA-Eur ecosystem.

Potential application in EAGLE: VeraGrid could be used as an alternative or complementary transmission analysis framework for EAGLE. Its Python engine, graphical interface, contingency-analysis capabilities and support for PyPSA, CGMES and MATPOWER formats make it particularly interesting for inspecting and independently validating the Spanish network model and for analysing transmission constraints and transfer capacities.

2. Data Sources for the EAGLE Model

2.1 Data Sources Overview

Table 2.1: Comparison of data sources for EAGLE requirements

EAGLE Requirement	REData	CNMC	PyPSA-Eur	REE Access Capacity	ENTSO-E
Historical demand	Direct	No	Time series	No	Historical load
Historical generation	Direct	No	Generation datasets	No	Actual generation
Energy mix	Direct	No	Modelled generation mix	No	By production type
Grid location	Limited	Access nodes / maps	Georeferenced network	Transmission nodes	Dataset-dependent
Substations / buses	No	Access nodes / substations	Yes	Transmission nodes	Grid-model dependent
Transmission lines	No	No	Yes	No	Grid datasets / CGMES
Connected topology	No	No	Yes	No	Grid-model dependent
Line voltage	No	Node voltage levels	Yes	Node voltage	Grid-model dependent
Circuits / conductors	No	No	OSM-derived / processed	No	Grid-model dependent
R / X parameters	No	No	Available / partly derived*	No	Possible in detailed CGMES models
Line capacity	No	No**	Modelled / estimated	No***	Dataset-dependent
Access capacity	No	Direct	No	Direct, node-level	Not primary purpose
HVDC	Operational information	Not a network dataset	Yes	No	Interconnection / grid data
Coordinates / GIS	Limited	Interactive maps	Yes	Node identification	Dataset-dependent
Spatial / nodal loads	No	No	Spatially allocated / modelled	No	Mostly aggregated
Time-series demand	Yes	No	Yes	No	Yes
Renewable availability	Historical generation	No	Availability time series	No	Generation / forecasts
Power plants / generators	Aggregated data	No	Yes	No	Mostly aggregated data
Cross-border flows	Spanish exchanges	No	Modelled interconnections	No	Direct
Simulation-ready model	No	No	PyPSA network	No	Detailed grid models may require conversion
CIM / CGMES	No	No	Not native source format	No	Core grid-exchange standard
Automation	REST API	Maps / operator data	Automated workflow	CSV / XLSX processing	REST API
Downloadable / machine-readable data	Yes	Depends on operator	Yes	CSV / XLSX	Yes
Open / reproducible workflow	Open data / API	Public data	Yes	Public official data	Depends on dataset****
Spain coverage	Yes	Yes	Yes	Yes	Yes
Main value for EAGLE	**Official Spanish operational time series**	**Centralised T+D access-capacity reference**	**Simulation-ready Spanish HV base model**	**Official TSO nodal access capacity**	**European operational and validation data**

* – PyPSA-Eur provides resistance, reactance and other electrical parameters in the processed network model. However, some values are derived from reconstructed network information and standard line types rather than official TSO asset data.

** CNMC access capacity should not be confused with the thermal transmission capacity of an individual power line.

*** REE access capacity represents the capability of a transmission-network node to accommodate additional demand or generation under the applicable access criteria. It should not be interpreted as the thermal rating of

2.2 Data Source Descriptions

2.2.1 REE – REData

Source: Red Eléctrica de España (Redeia).

Access: https://www.ree.es/es/datos/apidatos

Description: Official data platform of the Spanish electricity system providing historical and operational information through a REST API.

Relevant data:

- Electricity demand.
- Generation by technology.
- Generation mix.
- Installed generation capacity.
- Renewable generation and capacity.
- Electricity balance.

Spatial coverage: Spanish electricity system, including geographical disaggregation for different electrical systems and regions depending on the dataset.

Temporal coverage and resolution: Historical time series with hourly, daily, monthly or annual aggregation depending on the dataset.

Data format: JSON through the API. Some datasets can also be downloaded through the web interface.

Access method: REST API and REData web interface.

Processing required: API queries, JSON parsing, temporal filtering, data cleaning and transformation into tabular time series suitable for the EAGLE model.

Main limitations:

- Does not provide a detailed electrical topology of the Spanish transmission network.
- Does not provide the electrical parameters required to model individual transmission lines.
- Most information is aggregated rather than nodal.

Application in EAGLE: Construction of realistic historical demand and generation time series and validation of operational and future scenarios simulated with the EAGLE model.

2.2.2 CNMC – Electricity Network Access Capacity Maps

Source: Comisión Nacional de los Mercados y la Competencia (CNMC).

Access: https://www.cnmc.es/prensa/mapas-capacidad-redes-electricas-20260415

Description: Official CNMC platform integrating information on available electricity network

access capacity published by Spanish transmission and distribution system operators. Separate interactive maps are available for generation and demand access capacity.

Relevant data:

- Electricity network nodes and substations.
- Available generation access capacity.
- Available demand access capacity.
- Transmission or distribution system operator.
- Province.
- Voltage level.
- Number of available positions at substations.
- Generation access capacity differentiated by technology for the transmission network.
- Demand access capacity according to the applicable evaluation criteria.

Spatial coverage: Spanish electricity transmission and distribution networks.

Temporal coverage and resolution: Current network access-capacity information updated monthly by the corresponding transmission and distribution system operators.

Data format: Interactive geospatial maps integrating information supplied by the Spanish electricity network operators. The underlying publication formats depend on the corresponding network operator.

Access method: CNMC interactive generation and demand access-capacity maps. The original information can also be consulted through the corresponding transmission and distribution system operators.

Processing required: Identification of relevant nodes and voltage levels, extraction of access-capacity information and geographical/electrical matching with the nodes of the EAGLE network model.

Main limitations:

- Access capacity is not equivalent to the thermal capacity or rating of a transmission line.
- The platform does not constitute a complete electrical network model.
- It does not provide the complete set of line impedance parameters required for power-flow simulation.
- Data originate from multiple transmission and distribution system operators and may require additional harmonisation.
- Available access capacities should not necessarily be treated as simultaneously available or directly additive between nodes.

Application in EAGLE: Provides an official and frequently updated reference for the available

generation and demand access capacity at Spanish electricity network nodes. Particularly relevant for validating or complementing the hosting-capacity indicators calculated by EAGLE and for identifying areas where new electricity demand may face connection constraints.

2.2.3 PyPSA-Eur

Source: PyPSA-Eur project, currently maintained by the Department of Digital Transformation in Energy Systems at the Technical University of Berlin.

Access:

- Official PyPSA-Eur repository: https://github.com/PyPSA/pypsa-eur
- Latest PyPSA-Eur releases: https://github.com/PyPSA/pypsa-eur/releases
- Official PyPSA-Eur documentation: https://pypsa-eur.readthedocs.io/en/latest/
- Data sources and data versioning: https://pypsa-eur.readthedocs.io/en/latest/data_sources/
- PyPSA public data repository: https://data.pypsa.org/

Description: Open model dataset of the European energy system at transmission-network level. PyPSA-Eur combines multiple open data sources through a reproducible Snakemake workflow and produces a simulation-ready European electricity network compatible with the PyPSA modelling framework.

Relevant data:

- High-voltage electricity transmission network.
- Network buses and substations.
- AC transmission lines and cables.
- HVDC links.
- Geographical coordinates and network geometry.
- Voltage levels.
- Electrical line parameters.
- Transmission capacities.
- Existing power plants, including location, capacity and technology.
- Electricity demand time series.
- Spatial distribution of electricity demand.
- Renewable generation availability time series.
- Geographical wind and solar generation potentials.
- Additional energy-system data used for sector-coupled modelling.

Spatial coverage: European energy system covering the ENTSO-E area, including Spain. The default electricity network represents the high-voltage transmission system, while lower voltage thresholds can optionally be considered when constructing the network.

Temporal coverage and resolution: Depends on the individual dataset used by the PyPSA-Eur workflow. The model supports hourly time-series representation and combines network data with historical electricity demand, renewable-resource and energy-system datasets. Input datasets are independently versioned to ensure reproducibility.

Data format: Multiple machine-readable formats are used throughout the workflow, including CSV, GeoJSON and NetCDF. The final electricity network can be stored as a PyPSA network in NetCDF format.

Access method: Open-source PyPSA-Eur repository and automated data-retrieval workflow. Input datasets can be retrieved automatically according to the versions defined by PyPSA-Eur. Publicly redistributable datasets are also mirrored through the PyPSA data infrastructure.

Processing required: Configuration of the PyPSA-Eur workflow, selection of Spain as the geographical scope, retrieval of the required input datasets and construction of the electricity network. Additional validation and replacement of estimated parameters with official Spanish data should be performed where higher accuracy is required.

Main limitations:

- The transmission network is primarily reconstructed from OpenStreetMap rather than obtained directly from an official Spanish TSO electrical model.
- Some electrical network parameters and capacities are derived or estimated rather than obtained directly from official TSO data.
- Completeness and accuracy therefore depend partly on the underlying OpenStreetMap information and reconstruction procedure.
- Different input datasets have different reference years, licences and update frequencies.
- The European model may require Spanish-specific validation and enrichment for detailed nodal hosting-capacity studies.

Application in EAGLE: Strong candidate for the initial electrical-network dataset of the EAGLE model. PyPSA-Eur can provide a geographically referenced and simulation-ready representation of the Spanish high-voltage transmission network together with generation, demand and renewable-resource information. The Spanish subset can be used as the initial EAGLE network and subsequently validated, corrected and enriched with official Spanish sources such as REData, CNMC and Red Eléctrica.

2.2.4 REE – Transmission Network Access Capacity

Source: Red Eléctrica de España (Redeia), as Spanish Transmission System Operator (TSO).

Access:

- Demand access capacity: https://www.ree.es/es/clientes/consumidor/acceso-conexion/conoce-la-capacidad-de-acceso

- Generation access capacity: https://www.ree.es/es/clientes/generador/acceso-conexion/conoce-la-capacidad-de-acceso
- Status of demand access requests: https://www.ree.es/es/clientes/consumidor/acceso-conexion/conoce-el-estado-de-las-solicitudes

Description: Official information published by Red Eléctrica on the available and occupied access capacity at nodes of the Spanish transmission network. Separate datasets are published for electricity demand and generation. The information provides a direct reference for evaluating the capability of transmission-network nodes to accommodate additional consumption or generation.

Relevant data:

- Transmission-network nodes.
- Voltage level of the connection node.
- Available demand access capacity.
- Occupied demand access capacity.
- Available generation access capacity.
- Occupied generation access capacity.
- Capacity associated with existing access permissions.
- Capacity reserved for demand connected through distribution networks.
- Capacity associated with storage where applicable.
- Capacity-sharing zones between different substations.
- Information on dedicated positions for demand.
- Nodes subject to specific access-capacity conditions or allocation procedures.

Spatial coverage: Nodes of the Spanish electricity transmission network managed by Red Eléctrica. The datasets provide node-level information and are therefore considerably more spatially detailed than national or regional aggregated electricity-system statistics.

Temporal coverage and resolution: Current access-capacity information rather than historical electrical operating time series. The published access capacities are periodically updated by Red Eléctrica. At the time of this assessment, the available demand and generation capacity datasets are updated monthly.

Data format: Access-capacity results are published in machine-readable and human-readable formats, including CSV, XLSX and PDF. Additional documentation describing the calculation criteria and interpretation of the published fields is also available.

Access method: Direct download from the Red Eléctrica access-and-connection website. CSV and XLSX files can be processed automatically and linked with the corresponding transmission-network nodes of the EAGLE model.

Processing required: Download and parsing of the published files, identification of the corre-

sponding transmission-network nodes, normalisation of node names and voltage levels, and geographical/electrical matching with the buses of the EAGLE network model. Particular attention should be given to the definitions of available, occupied and reserved access capacity.

Main limitations:

- Access capacity is not equivalent to the thermal rating of an individual transmission line.
- Published capacity represents the result of an access-capacity assessment considering several network criteria and should not be interpreted as a simple physical line limit.
- Available capacity values at different nodes should not automatically be assumed to be simultaneously available or directly additive.
- The published information does not constitute a complete electrical transmission-network model.
- Detailed line impedance parameters required for independent power-flow calculations are not provided through this dataset.
- Published values have an informative character and do not replace the specific access study performed for an individual connection request.

Application in EAGLE: This dataset is particularly relevant for EAGLE because the project aims to evaluate the capability of the electricity network to accommodate new large electricity consumers such as data centres. The official demand access capacity published for transmission-network nodes can be geographically and electrically matched with the corresponding buses of the EAGLE model.

It can therefore be used both as a direct grid-availability indicator for SiteOptimizer and as an official reference for validating the nodal hosting-capacity indicators calculated independently using the EAGLE electrical-network model.

2.2.5 ENTSO-E – Transparency Platform and Grid Data

Source: European Network of Transmission System Operators for Electricity (ENTSO-E).

Access:

- ENTSO-E Transparency Platform: https://transparency.entsoe.eu/
- Transparency Platform information: https://www.entsoe.eu/data/transparency-platform/
- ENTSO-E statistics and grid datasets: https://www.entsoe.eu/publications/statistics-and-data/
- CIM / CGMES grid-model information: https://www.entsoe.eu/data/cim/cim-for-grid-models-ex

Description: ENTSO-E provides harmonised European electricity-system information through several platforms and datasets. The Transparency Platform publishes operational and market-related information supplied by European transmission system operators, while additional ENTSO-E resources provide transmission-network datasets and standardised Common Information Model (CIM) / Common Grid Model Exchange Specification (CGMES) information.

Relevant data:

- Actual electricity load.
- Day-ahead and longer-term load forecasts.
- Actual generation.
- Generation by production type.
- Installed generation capacity.
- Wind and solar generation forecasts.
- Cross-border electricity flows.
- Transmission information.
- Transmission and generation outages.
- Balancing information.
- European bidding-zone and system-level information.
- European transmission-grid models for specific studies, subject to dataset-specific access conditions.
- CIM / CGMES specifications and reference information for exchanging detailed transmission-network models.

Spatial coverage: European electricity system covering the ENTSO-E area, including Spain and its electrical interconnections with neighbouring systems. The spatial resolution depends strongly on the specific dataset: Transparency Platform information is commonly provided at bidding-zone or system level, whereas dedicated grid-model datasets can contain considerably more detailed transmission-network information.

Temporal coverage and resolution: Depends on the dataset. The Transparency Platform provides historical, current and forecast electricity-system information with resolutions appropriate to the corresponding load, generation, transmission, balancing or outage dataset.

Data format: Transparency Platform information can be downloaded in machine-readable formats including XML, CSV and XLSX depending on the access method and dataset. Detailed grid models are increasingly exchanged using CIM / CGMES-based representations.

Access method: Data can be consulted and downloaded through the ENTSO-E Transparency Platform. Programmatic access is available through the ENTSO-E Transparency Platform REST API for supported datasets.

Detailed grid-model datasets have different access conditions. Some datasets are available to universities, companies and other institutions upon request, while access to more advanced network datasets may require additional agreements or a Non-Disclosure Agreement.

Processing required: Selection of the relevant Spanish bidding zone or network dataset, automated data retrieval where appropriate, parsing of the corresponding XML/CSV/XLSX

or CGMES representation, temporal harmonisation and conversion into the internal data structures used by the EAGLE electrical model.

Main limitations:

- Transparency Platform data should not be interpreted as a directly downloadable complete electrical model of the Spanish transmission network.
- Much of the operational information is aggregated at bidding-zone or system level rather than individual electrical nodes.
- Detailed European grid datasets have dataset-specific access conditions and are not necessarily fully open for unrestricted download.
- Access to some advanced network datasets may require a formal request and a Non-Disclosure Agreement.
- The spatial and temporal resolution varies considerably between datasets.
- CGMES models require specialised processing before they can be directly used in the selected EAGLE simulation framework.

Application in EAGLE: ENTSO-E can provide an independent European reference for validating Spanish demand, generation, renewable production and cross-border electricity exchanges. This is particularly useful for validating the boundary conditions of the Spanish EAGLE model and its electrical interconnections with neighbouring countries.

ENTSO-E is also relevant as a potential source of more detailed transmission-network models in CIM/CGMES format if suitable datasets can be obtained. Such models could be compared with the reconstructed PyPSA-Eur Spanish network and used to validate or improve its topology and electrical parameters.

3. Power Grid Reconstruction Methodologies

3.1 Xiong et al. – OSM/PyPSA

Reference: Xiong et al., "Modelling the high-voltage grid using open data for Europe and beyond".

Objective: Develop an open and reproducible methodology for constructing a topologically connected and simulation-ready representation of the European high-voltage transmission grid using primarily OpenStreetMap data. The methodology is designed to overcome the lack of openly available detailed transmission-system models and is integrated into the PyPSA-Eur workflow.

Input data:

- OpenStreetMap AC transmission lines and cables.
- OpenStreetMap substations.
- OpenStreetMap HVDC connections.
- Geographical coordinates and geometries.
- Voltage levels.
- Number of circuits and cables when available.
- Line frequency information.
- Standard electrical line types from power-system modelling libraries.

Methodology:

- Retrieval of power-system infrastructure from OpenStreetMap using the Overpass API.
- Extraction of AC lines, cables, substations and HVDC infrastructure.
- Cleaning and normalisation of incomplete or inconsistent OSM attributes.
- Selection of high-voltage substations and transmission infrastructure.
- Identification of line endpoints and conversion of geographical infrastructure into network buses and branches.
- Detection of transmission lines passing through substations and splitting of lines where necessary.
- Spatial clustering of nearby buses belonging to the same substation.
- Connection of different voltage levels within substations using transformers.
- Reconstruction of a topologically connected transmission network.

- Separate processing and integration of HVDC connections.
- Assignment of standard electrical line types according to voltage and available OSM characteristics.
- Estimation of missing electrical parameters and transmission capacities from standard line types, line length and circuit information.
- Conversion of the resulting network into a representation directly usable by PyPSA and PyPSA-Eur.

Inferred / generated information:

- Network buses.
- Connected AC transmission lines and cables.
- HVDC links.
- Transformers between voltage levels.
- Connected high-voltage network topology.
- Estimated resistance and reactance.
- Estimated transmission capacities.
- Electrical connections not explicitly represented in the original geographical data.
- Simulation-ready PyPSA network representation.

Output: An open European high-voltage transmission network containing buses, AC lines and cables, HVDC links and transformers. The processed network can be exported as tabular datasets such as CSV files and used directly within the PyPSA/PyPSA-Eur modelling ecosystem.

Validation: The reconstructed network is compared with ENTSO-E transmission-network statistics and other available official data. The study reports strong agreement in total transmission-route and circuit lengths across European countries. Electrical line parameters are additionally compared with an official German transmission-system model, and the reconstructed network is tested in a full PyPSA-Eur optimisation covering 8760 hours. The study also investigates discrepancies in Spain and compares the reconstructed topology with official Spanish geographical information.

Main assumptions:

- Missing circuit information may be completed using predefined assumptions.
- Lines are assigned standard electrical types according to voltage and available characteristics.
- Electrical parameters such as resistance and reactance are derived from standard line types when official parameters are not available.
- Transmission capacities are estimated from the assigned technical characteristics rather than obtained directly from the TSO.

- Nearby buses can be clustered when they are considered to represent the same physical substation.
- Transformers can be synthetically introduced when buses at different voltage levels belong to the same substation.

Main limitations:

- The reconstructed grid is not an exact representation of the official transmission network operated by each TSO.
- Results depend on the completeness and quality of OpenStreetMap data.
- Some electrical parameters are inferred from standard equipment types rather than obtained from official network data.
- Transmission capacities are estimates and should not automatically be interpreted as official operational limits.
- Some transformers are synthetic because sufficiently complete transformer information is not available from OSM.
- The methodology mainly focuses on high-voltage transmission infrastructure of 200 kV and above.

Application in EAGLE: This methodology is particularly relevant for constructing the base electrical topology of the EAGLE Spanish transmission-network model. Its workflow can be used to transform raw OpenStreetMap geographical information into connected buses, transmission lines, transformers and HVDC links. The proposed use of standard line types provides a practical method for estimating missing resistance, reactance and transmission capacities when official values are unavailable. In addition, the resulting network is directly compatible with PyPSA/PyPSA-Eur, making the methodology especially suitable if PyPSA is selected as one of the EAGLE simulation and optimisation frameworks.

3.2 Rivera et al. – OSM/Transnet

Reference: Rivera et al., "Automatic Generation of Real Power Transmission Grid Models From Crowdsourced Data", IEEE Transactions on Smart Grid, 2019.

Objective: Automatically reconstruct transmission grid models from incomplete and unstructured crowdsourced OpenStreetMap (OSM) data. The methodology aims to infer electrical connections between power lines, substations and generators and transform the resulting topology into a simulation-ready electrical network model.

Input data:

- OpenStreetMap power lines.
- Substations.
- Power plants and generators.
- Geographical coordinates and geometries.
- Voltage levels and other available OSM attributes.

- Population data for load estimation.
- Average electricity consumption data for load estimation.

Methodology:

- Extraction of power-related objects from OpenStreetMap.
- Identification of relevant transmission voltage levels.
- Filtering of transmission lines and possible network endpoints for each voltage level.
- Spatial identification of intersections between power lines, substations, power plants and generators.
- Recursive traversal of connected lines to identify the endpoints of electrical circuits.
- Automatic inference of power circuit relations that are not explicitly defined in OpenStreetMap.
- Treatment of T-junctions by dividing them into individual electrical relations.
- Construction of a connected transmission network topology.
- Estimation of missing electrical parameters.
- Estimation of nodal loads using geographical population information and electricity consumption.
- Conversion of the reconstructed network into Common Information Model (CIM).
- Conversion of CIM models into MATLAB/Simulink models for power-flow simulation.

Inferred / generated information:

- Connected transmission network topology.
- Electrical circuit relations between network elements.
- Connections between lines, substations and generators.
- Estimated line resistance and reactance.
- Estimated nodal loads.
- CIM representation of the reconstructed network.
- Simulation-ready MATLAB/Simulink network model.

Output: Reconstructed transmission network models that can be stored in JSON and CSV formats and subsequently converted to CIM and MATLAB/Simulink representations suitable for power-flow simulations.

Validation: The methodology was evaluated for 14 countries by comparing the automatically inferred grid with manually defined OSM circuit relations and official transmission-network statistics. The topology-inference algorithm achieved high recall when compared with manually defined OSM relations. For Spain, the reconstructed 220–400 kV network rep-

resented approximately 63% of the officially reported transmission-line length using the OSM data considered in the study.

Main assumptions:

- Only transmission voltage levels with sufficient representation in OSM are considered.
- Missing conductor characteristics are inferred from technical assumptions associated with voltage levels.
- Line resistance and reactance are approximated from line length and assumed conductor characteristics.
- Electricity demand is geographically distributed using population information and average electricity consumption.
- The network is assumed to be a balanced three-phase system for the demonstrated power-flow simulations.

Main limitations:

- The quality of the reconstructed network depends strongly on the completeness and accuracy of OpenStreetMap.
- The Spanish evaluation is based on old OSM data and should not be interpreted as a current representation of the Spanish grid.
- The reported 63% value for Spain is primarily a comparison of transmission-line length and does not represent 63% accuracy of all electrical parameters or network topology.
- Resistance and reactance are rough estimates rather than official transmission-system parameters.
- Nodal loads are estimated from population data rather than obtained from actual electrical measurements.
- Intentional network loops can be ignored by the proposed inference algorithm to reduce computational complexity.

Application in EAGLE: The main value for EAGLE is methodological rather than as a direct source of Spanish transmission-grid data. The proposed spatial connectivity inference can be adapted to reconstruct electrical relations when OpenStreetMap data do not explicitly define them. Its approaches for estimating missing network parameters and nodal loads also provide possible strategies for completing incomplete EAGLE datasets. The CIM conversion demonstrates an additional method for transforming the reconstructed topology into a standard electrical network representation.

4. Data Center Planning Papers

4.1 Scientific Literature

Paper / Reference	Main Contribution	Summary Section
Location Selection Methodology for Data Center with Renewable Energy Integration Ayyildiz et al. (2025) *Renewable Energy* DOI: 10.1016/j.renene.2025.123270	**Multi-Criteria Site Selection** • Site selection methodology • Renewable energy integration • Location evaluation criteria	4.2.1
Geospatial Suitability Analysis for Data Center Placement: A Case Study in Texas, USA Arzumanyan et al. (2025) *Sustainable Cities and Society* DOI: 10.1016/j.scs.2025.106687	**Geospatial Suitability** • Geographic site assessment • Infrastructure availability • Spatial suitability criteria	4.2.2
Optimal County-Level Siting of Data Centers in the United States Vabson et al. (2026) *arXiv Preprint* arXiv: 2601.16315	**Geographical Site Selection** • Electricity and grid connection costs • Renewable energy potential • Water and cooling requirements • Fiber-optic infrastructure • Geographic cost optimization	4.2.3

4.2 Literature Review: Data Center Location Optimization

4.2.1 Location Selection Methodology for Data Center with Renewable Energy Integration

Authors: Ertugrul Ayyildiz, Betul Yildirim and Nezir Aydin (2025).

Publication: Renewable Energy.

Summary and Introduction

This paper addresses the increasing energy demand and environmental impact of data centers by proposing a Multi-Criteria Decision-Making (MCDM) framework for selecting optimal locations that integrate renewable energy sources.

The research aims to address the uncertainty and imprecision inherent in expert evaluations through the use of Picture Fuzzy Sets (PiFS), which incorporate positive, neutral, negative and refusal membership degrees.

Methodology

- **Hierarchical Criteria Framework:**

  Five main criteria and 34 subcriteria were defined:

  - Environmental: 26.71%.
  - Renewable Energy: 25.00%.
  - Accessibility and Infrastructure: 18.00%.
  - Economic: 17.00%.
  - Social and Governance: 13.00%.

- **Criteria Weighting (PiF-SWARA):**

  The Picture Fuzzy Step-wise Weight Assessment Ratio Analysis (PiF-SWARA) method was applied using linguistic evaluations collected from seven industry experts.

- **Alternative Evaluation (PiF-VIKOR):**

  The Picture Fuzzy VIKOR method was used to evaluate and rank six candidate cities in Turkey:

  - Konya.
  - İzmir.
  - Trabzon.
  - Antalya.
  - Şanlıurfa.
  - Çanakkale.

  The alternatives were assessed according to their distance from ideal solutions.

- **Sensitivity and Comparative Analysis:**

  The decision strategy parameter τ was varied within the interval:

  $$\tau \in [0.1, 0.9]$$

  The resulting rankings were compared with traditional MCDM methods, including TOPSIS, SAW, WASPAS and WISP.

Results

- **Most Critical Subcriteria:**

  The most influential individual factors were:

  - Seismic intensity (C13): $w = 0.0588$.
  - Water resource availability (C11): $w = 0.0580$.

  - Carbon emissions (C15): $w = 0.0537$.
  - Average annual sunshine duration (C51): $w = 0.0536$.

- **Final Ranking:**

  İzmir achieved the highest overall ranking, with:

$$Q_{\text{Izmir}} = 0.017$$

  followed by Konya:

$$Q_{\text{Konya}} = 0.396$$

  The remaining alternatives were ranked as follows:

$$\text{Antalya} > \text{Sanliurfa} > \text{Trabzon} > \text{Canakkale}$$

- **Advantages of İzmir:**

  İzmir stood out due to its combination of telecommunications infrastructure, relatively low seismic vulnerability and significant wind and solar energy potential.

Conclusions

The study demonstrates that incorporating renewable energy and environmental safety criteria significantly changes traditional data center location priorities.

The hybrid PiF-SWARA–PiF-VIKOR framework showed mathematical robustness against variations in expert preferences, supporting its applicability to complex infrastructure location decisions.

4.2.2 Geospatial Suitability Analysis for Data Center Placement: A Case Study in Texas, USA

Authors: Mariam Arzumanyan, Edna Rodriguez Calzado, Ning Lin, Vaibhav Bahadur, Jani Das, Tingwei Lucy Ko and Lars Koesterke (2025).

Publication: Sustainable Cities and Society.

Summary and Introduction

This study develops a regional geospatial suitability analysis framework to evaluate potential data center locations across the state of Texas, USA.

The proposed methodology combines Machine Learning (ML) algorithms with Multi-Criteria Decision-Making (MCDM) techniques within a Geographic Information System (GIS).

The objective is to balance energy demand, climate-related risks and infrastructure availability when identifying suitable locations for new data centers.

Methodology

- **Spatial Variables:**

  Five main dimensions were considered:

  1. **Energy Access:** Current and projected electrical capacity and distance to high-voltage transmission lines.
  2. **Natural Hazards:** FEMA datasets covering flooding, cold waves, hurricanes, tornadoes and wildfires.
  3. **Skilled Workforce:** Employment information based on NAICS code 518210 and IMPLAN datasets.
  4. **Fiber-Optic Connectivity:** Telecommunications infrastructure information obtained from GeoTel.
  5. **Land Use and Availability:** Land-cover information from the USGS National Land Cover Database (NLCD).

- **Weight Determination Using Random Forest:**

  A Random Forest (RF) classification model was trained using:

  – 800 existing data center locations ($y = 1$).

  – 1,000 randomly generated control locations ($y = 0$).

  Feature importance scores derived from the reduction in Gini impurity were used to establish the relative weights of the suitability criteria.

- **Weighted Linear Combination (WLC):**

  The spatial variables were combined using a weighted linear suitability model:

$$S_i = \sum_{j=1}^{n} w_j x_{ij}$$

  where:

  – S_i is the suitability score of location i.

  – w_j is the weight assigned to criterion j.

  – x_{ij} is the normalized value of criterion j at location i.

  – n is the number of criteria.

- **Scenario Analysis:**

  Five evaluation scenarios were implemented:

  1. **Baseline Scenario:** Data-driven weighting based on Random Forest feature importance.
  2. **Energy Optimization Scenario:** 50% of the total weight assigned to energy-related

variables, targeting hyperscale and cloud data centers.

3. **Climate Risk Minimization Scenario:** 50% of the total weight assigned to natural hazards, targeting Tier III and Tier IV data centers.
4. **Neutral Scenario:** Equal weights assigned to all criteria.
5. **Random Sensitivity Scenario:** Dynamic variation of criteria weights to evaluate ranking stability.

Results

- **Key Influential Factors:**

  In the Random Forest baseline model, the most relevant predictors were:

  – Skilled workforce availability: 13%.

  – Current electrical capacity: 11%.

  – Distance to transmission lines: 11%.

  – Land cover: 11%.

- **Geographical Distribution:**

  The highest suitability scores were consistently concentrated around major metropolitan regions:

  – Dallas–Fort Worth.

  – Houston.

  – Austin.

  – San Antonio.

  – Permian Basin energy hub, particularly Odessa.

- **Scenario Trade-offs:**

  The Energy Optimization Scenario favored regions with high electricity generation capacity, particularly West Texas.

  Conversely, the Climate Risk Minimization Scenario penalized coastal areas along the Gulf of Mexico, including Houston and Corpus Christi, due to their exposure to hurricanes and flooding.

Conclusions

The study concludes that inland metropolitan regions provide a favorable balance between digital infrastructure, workforce availability and environmental resilience.

However, it emphasizes the need to decarbonize the regional electricity system operated by ERCOT and to incorporate long-term climate resilience into data center infrastructure planning.

4.2.3 Optimal County-Level Siting of Data Centers in the United States

Authors: Maria Vabson, Muhy Eddin Za'ter, Amir Sajadi, Kyri Baker and Bri-Mathias Hodge (2026).

Publication: Preprint / University of Colorado Boulder.

Summary and Introduction

This study formulates a quantitative Capacity Expansion Model (CEM) to determine optimal data center locations at the county level across the contiguous United States.

The analysis covers 3,066 counties and simultaneously considers electricity consumption, water requirements, fiber-optic connectivity and the possibility of colocating clean electricity generation and energy storage systems.

The main objective is to identify cost-effective locations while accounting for long-term infrastructure investment and environmental sustainability.

Methodology

- **Mathematical Formulation:**

  The location problem is formulated as a multi-objective Mixed-Integer Linear Programming (MILP) optimization model.

  The model is implemented using Pyomo and solved with the Gurobi optimization solver.

- **Energy and Water Demand Modeling:**

  Electricity and water consumption are estimated by combining IT equipment demand with:

  - Power Usage Effectiveness (PUE).
  - Water Usage Effectiveness (WUE).
  - Climate zones defined by the International Energy Conservation Code (IECC).

  The total data center electricity demand can be expressed as:

  $$P_{\mathrm{DC}} = P_{\mathrm{IT}} \cdot \mathrm{PUE}$$

  where P_{IT} represents IT equipment power demand.

- **Colocated Energy Technologies:**

  The model considers several electricity generation and storage technologies:

  - Solar photovoltaic generation.
  - Onshore wind generation.
  - Geothermal generation.
  - Small Modular Nuclear Reactors (SMRs).

  - Battery Energy Storage Systems (BESS).

- **Infrastructure Costs:**

  The optimization incorporates the costs associated with:

  - Electrical substations.
  - Transmission line construction and connection.
  - Fiber-optic infrastructure.
  - Energy generation and storage assets.

- **Objective Function:**

  The optimization minimizes total system costs, including annualized capital investment, fixed operating and maintenance costs, variable electricity and water costs, renewable energy curtailment costs and network charges.

  A general representation of the objective function is:

$$\begin{aligned} \min\ C_{\text{total}} = {} & C_{\text{CAPEX}} + C_{\text{O\&M}} \\ & + C_{\text{energy}} + C_{\text{water}} \\ & + C_{\text{curtailment}} + C_{\text{grid}} \end{aligned}$$

  where each term represents the corresponding cost component.

Results

- **Economic Drivers:**

  Initial capital investment and fixed operating costs showed the strongest positive correlation with total project costs.

- **Geographical Patterns:**

  Counties located in the southern and eastern regions of the United States generally exhibited lower total deployment costs, partly due to favorable local solar capacity factors.

- **Impact of Colocated Renewable Energy:**

  Allowing higher shares of directly integrated variable renewable generation, particularly wind and solar power combined with battery storage, increased clean electricity utilization.

  This integration also contributed to reducing long-term operating costs.

- **High-Cost Technologies:**

  Geothermal generation showed a strong positive correlation with total project costs due to its comparatively high initial investment requirements relative to solar and wind technologies.

Conclusions

The study quantitatively demonstrates that, although short-term data center location decisions are often driven by low electricity tariffs and land prices, long-term investments in colocated renewable generation and energy storage can provide a more cost-effective and environmentally sustainable strategy.

The results highlight the importance of jointly optimizing data center location, electricity infrastructure, renewable generation and storage capacity to accommodate the increasing computational demand associated with artificial intelligence and cloud computing.

4.3 Criterios para el Data Center Planning

1. **Impactos medioambientales**

   - Disponibilidad de recursos hídricos. (Riesgo de escasez de agua y presión sobre los recursos hídricos locales)
   - Disponibilidad y uso del suelo, y restricciones territoriales.
   - Necesidad de energia para la refrigeración (Clima local (temperatura y humedad) y posibilidades de *free cooling*.)

2. **Riesgos medioambientales y climaticos**

   - Exposición a fenómenos naturales (inundaciones, incendios, olas de frío, etc.).
   - Intensidad sísmica.

3. **Disponibilidad deEnergías renovables**

   - Potencial del viento.
   - Radiación solar anual y potencial de generación fotovoltaica.
   - Potencial de otras fuentes renovables (geotermia, hidroeléctrica y biomasa; energía undimotriz cuando proceda).
   - Proximidad y acceso a instalaciones de generación renovable.
   - Posibilidad de integrar generación renovable y almacenamiento energético *in situ*.
   - Variabilidad horaria de la producción renovable y complementariedad con el perfil de demanda.
   - Impacto ambiental de las instalaciones de generación renovable.

4. **Accesibilidad e infraestructura**

   - Proximidad a redes de telecomunicaciones.
   - Proximidad a redes de fibra óptica de alta capacidad y necesidades de ampliación.
   - Proximidad a líneas de alta tensión y subestaciones.
   - Congestión de la red. (Capacidad y demanda eléctrica.)
   - Perfiles de tensión y estabilidad nodal (asegurar que la incorporación del centro de datos no comprometa la seguridad ni la estabilidad de la red).
   - Nodos con margen de reserva (energía y capacidad de infraestructura).
   - Capacidad de generación eléctrica existente y planificada en la zona.
   - Capacidad de conexión y posibles necesidades de refuerzo o ampliación de la red eléctrica

y las subestaciones.
- Fiabilidad, continuidad y redundancia del suministro eléctrico.
- Acceso a carreteras principales, ferrocarril, aeropuertos y servicios logísticos.
- Disponibilidad de suelo adecuado y espacio para futuras ampliaciones.

5. **Criterios económicos**

- Costes del terreno.
- Inversión inicial.
- Incentivos financieros.
- Precio de la electricidad y costes energéticos durante la operación.
- Costes de refrigeración y consumo de agua.
- Costes de conexión y ampliación de las redes eléctrica y de telecomunicaciones.
- Costes de inversión, operación y mantenimiento de la generación y del almacenamiento energético *in situ*.
- Coste total a lo largo de la vida útil del proyecto, considerando el horizonte de inversión y la financiación.

6. **Criterios sociales y políticos**

- Conflictos con reservas y parques naturales.
- Entorno regulatorio.
- Aceptación comunitaria.
- Disponibilidad de personal técnico cualificado.
- Seguridad física del emplazamiento y protección frente a amenazas.
- Compatibilidad con la planificación territorial y los usos del suelo.
- Incentivos y políticas locales de promoción de inversiones y desarrollo de infraestructuras.